CompTIA Security+ Exam SY0-701

Lesson 11

Enhance Application Security Capabilities

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Lesson 11

Topic 11A

Application Protocol Security Baselines

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Secure Protocols

- Many of the protocols used today were developed many decades ago
  - Functionality was primary focus
  - Trustworthiness was assumed
  - Cybersecurity was less of an issue than it is today
- Insecure Protocols
  - Transmit data in clear text format
  - Generally, cannot be secured
  - Must be avoided
- Secure Protocols
  - Same functionality and secure
  - More complex to configure

Insecure	Secure Alternative
Telnet	SSH
HTTP	HTTPS
FTP	FTPS/SFTP

Secure Directory Services

- A Network directory contains
  - Subjects (users, computers, and services)
  - Objects (directories and files) available in the environment
  - Permissions that subjects have over objects
  - High-value attack target

- Lightweight Directory Access Protocol (LDAP)
  - Default is cleartext communication

Simple Network Management Protocol Security

- Simple Network Management Protocol (SNMP)
- Management and monitoring
- SNMP monitor + agents
- Provides very detailed information about systems
- Uses “Community Strings” default “Public” and “Private”
- Can be used to issue commands
- SNMPv3 has secure features, other versions should be avoided

File Transfer Services

- File Transfer Protocol
  - Cleartext
  - Used to host and share files
- SSH
  - Primarily used to access a shell remotely
  - Very versatile protocol
  - Can be used as a tunnel for other protocols
- FTP (SFTP) and FTP Over SSL (FTPS)
  - SFTP is FTP tunneled through SSH
  - FTPS is FTP secured using TLS

Email Services

```
  GNU nano 2.2.2          File: /etc/dovecot/dovecot.conf                Modified

protocols = imap imaps
#protocols = none

# A space separated list of IP or host addresses where to listen in for
# connections. "*" listens in all IPv4 interfaces. "[::]" listens in all IPv6
# interfaces. Use "*, [::]" for listening both IPv4 and IPv6.
#
# If you want to specify ports for each service, you will need to configure
# these settings inside the protocol imap/pop3/managesieve { ... } section,
# so you can specify different ports for IMAP/POP3/MANAGESIEVE. For example:
   protocol imap {
     listen = *:143
     ssl_listen = *:943
    }
#   protocol pop3 {
#     listen = *:10100
#     ..
#   }
#   protocol managesieve {
#     listen = *:12000
#     ..
#   }
#listen = *

# Disable LOGIN command and all other plaintext authentications unless
                              [ Read 1280 lines ]
^G Get Help  ^O WriteOut  ^R Read File ^Y Prev Page ^K Cut Text  ^C Cur Pos
^X Exit      ^J Justify   ^W Where Is  ^V Next Page ^U UnCut Text^T To Spell
```

Configuring mailbox access protocols on a server.

- SMTP
  - Cleartext by default
  - Transmit email between systems
  - SMTPS is secure configuration
- Open Relay
  - Improperly configured SMTP server
  - Used to send SPAM
- POP & IMAP
  - Used to access mailboxes
  - Cleartext by default
  - POPS & IMAPS are secure

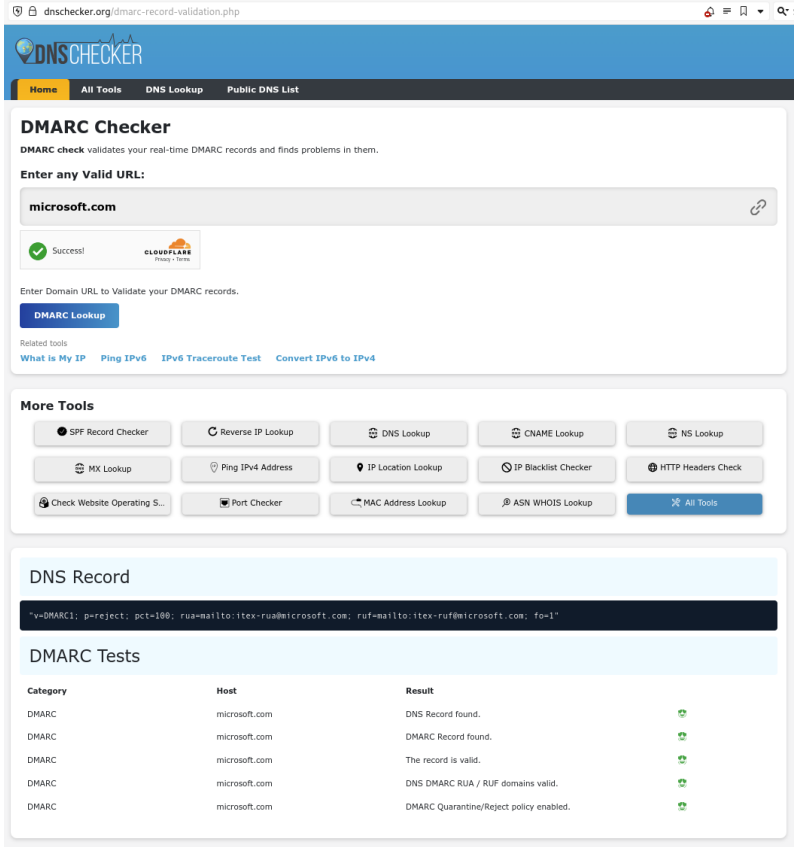
Email Security

- Sender Policy Framework (SPF)
  - Email validation method that helps detect and prevent sender address forgery
  - Uses data saved in DNS TXT Records
  - Can use to identify “authorized senders”
    - Hosted email
    - Marketing campaigns, etc.

```
└─$ dig txt microsoft.com
;; Truncated, retrying in TCP mode.

; <<>> DiG 9.18.12-1-Debian <<>> txt microsoft.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 55789
;; flags: qr rd ra; QUERY: 1, ANSWER: 16, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;microsoft.com.                 IN      TXT

;; ANSWER SECTION:
microsoft.com.          1590    IN      TXT     "fg2t@gov9424p2tdcuo94goe9j"
microsoft.com.          1590    IN      TXT     "t7sebee51jrj7vm932k531hipa"
microsoft.com.          1590    IN      TXT     "google-site-verification=M--CVfn_YwsV-2FGbCp_HFaEj23BmT0cTF4l8hXgpvM"
microsoft.com.          1590    IN      TXT     "google-site-verification=GfDnTUdATPsK1230J0mXbfsYw-3A9BVMVaKSd4DcKgI"
microsoft.com.          1590    IN      TXT     "d365mktkey=SxDf1EZxLvMwx6eEZUxzjFFgHoapF8DvtWEUjwq7ZTwx"
microsoft.com.          1590    IN      TXT     "hubspot-developer-verification=OTQ5NGIwYWEtODNmZi00YWE1LTkyNmQtNDhjMDMxY2JjNDAx"
microsoft.com.          1590    IN      TXT     "d365mktkey=QDa792dLCZhvaAOOCe2Hz6WTzmTssOp1snABhxWibhMx"
microsoft.com.          1590    IN      TXT     "d365mktkey=6358r1b7e13hox60tl1uagv14"
microsoft.com.          1590    IN      TXT     "google-site-verification=uFg3wr5PWsK8lV029RoXX8BUW0_E6qf1WEWVHhetkOY"
microsoft.com.          1590    IN      TXT     "docusign=d5a3737c-c23c-4bd0-9095-d2ff621f2840"
microsoft.com.          1590    IN      TXT     "d365mktkey=12gNNz9BHdaa3ZKZHB+64daJZxEWsFa0dxDeilxDoYYx"
microsoft.com.          1590    IN      TXT     "v=spf1 include:_spf-a.microsoft.com include:_spf-b.microsoft.com include:_spf-c.microsoft.com include:_spf-ssg-a.msft.net include:
spf-a.hotmail.com include:_spf1-meo.microsoft.com -all"
microsoft.com.          1590    IN      TXT     "8RPOXJ8zB59tu7PbysU7qCACrwXPoDVBZtLfthTnC4y9VJfLd841t5sQlEIfgSLJ4KOIA8pBzxmyvPuJuUvhOg=="
microsoft.com.          1590    IN      TXT     "d365mktkey=3uc1cf82cpv75012k70v9bvf2"
microsoft.com.          1590    IN      TXT     "facebook-domain-verification=fwzwhbbzwmg5fzgotc2go51olc3566"
microsoft.com.          1590    IN      TXT     "google-site-verification=pjPOauSPcrfXOZS9jnPPa5axowcHGCDAli_86dCqFpk"
```

Displaying the TXT records for microsoft.com using the dig tool. (Screenshot used with permission from Microsoft.)

Email Security



Performing a DMARC lookup using the DNSChecker website https://dnschecker.org.

- DomainKeys Identified Mail (DKIM)
  - Sender signs emails using a digital signature
  - Receiver uses a DKIM record in the sender's DNS to verify the signature
- Domain-based Message Authentication, Reporting & Conformance (DMARC)
  - Uses the results of SPF and DKIM checks to define rules for handling messages
  - Provides reporting capabilities
    - Email activity
    - Identify systems sending emails
    - Identify unauthorized activity

Email Security

- Email Gateway
  - Control point for all incoming and outgoing email
  - Anti-spam filters and antivirus scanners
  - Sophisticated threat detection algorithms
    - Identify phishing attempts, Business Email Compromise (BEC) Attack
  - Harmful attachments and malicious URLs
    - URL Sanitization/Link Anonymization/Safe Linking/Web Link Transformation
- Secure/Multipurpose Internet Mail Extensions (S/MIME)
  - Encrypts emails to provide the confidentiality and integrity protections
  - Requires Public Key Infrastructure (PKI)

DNS Filtering

- Block or allow access to specific websites
  - DNS filter checks requests against a database of domain names
  - Block access to malicious sites
  - Content/Site Restrictions
  - Ad-blocking (Pi-Hole, AdGuard)
- OpenDNS opendns.com
- Quad9 quad9.net
- CleanBrowsing cleanbrowsing.org
- Cisco Umbrella umbrella.cisco.com/products/dns-layer-network-security
- CloudFlare DNS cloudflare.com/application-services/products/dns/

The Pi-hole administrative dashboard showing DNS resolution statistics. (Screenshot courtesy of Pi-hole.)

DNS Security

- DNS Contains valuable information about hosts on a network
- Internal records should not be accessible from the Internet
- DNS protocol is often exploited to perform data exfiltration
- DNS can be exploited to provide malicious data (ex. Attacker IP instead of real IP)

- DNS Security Extensions (DNSSEC)
  - Mitigate spoofing and poisoning attacks
  - Provides a validation process for DNS responses
  - Authoritative server for the zone creates a "package" of resource records (RRset)

Review Activity: Application Protocol Security Baselines

- Secure Protocols
- Transport Layer Security
- Secure Directory Services
- Simple Network Management Protocol Security
- File Transfer Services
- Email Services
- Email Security
- Email Data Loss Prevention
- DNS Filtering

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Topic 11B

Cloud and Web Application Security Concepts

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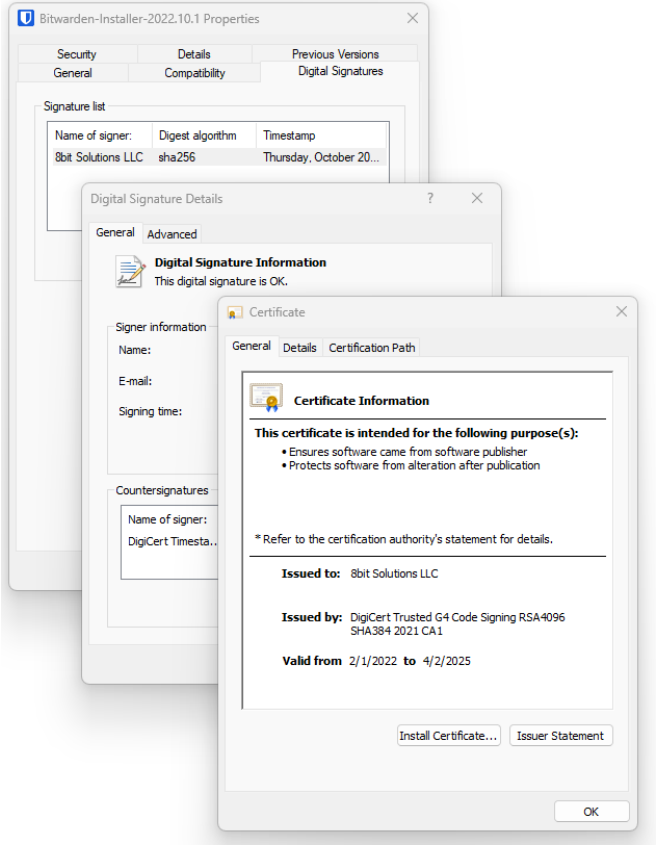
Secure Coding Techniques

- Pressure to release an application often overshadows the requirement to ensure it is secure
- Coding practices must implement secure development practices
- Code Signing
- Secure Cookies
- Static/Dynamic Code Analysis
- Peer Review



Reviewing the digital signature contained within the Bitwarden Password Management app installer.

Secure Coding Techniques

- Input Validation
- Attacker provides specially crafted data to an application to manipulate its behavior
- Injection Attack
- Methods used to perform input validation:
  - Allow/Block Lists
  - Data Type checks
  - Range checks
  - Regular Expressions
  - Encoding

Application Protections

- Data exposure
- Allows privileged information to be read by unauthorized user
  - Access token
  - Password
  - Personal data
- Error Handling
  - Safely handle and control errors
  - Report errors to logs instead of user interface
- Application Security in the Cloud
  - Application security supports the shared responsibility model
  - Secure applications running on a secure cloud platform

Review Activity:

- Secure Coding Techniques
- Application Protections
- Software Sandboxing

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Summary

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